

Horizon matching on real DROID recordings

Validation only · 12 full 30-epoch runs · selected epoch 1 for every model

a Match the training horizon

h10-trained – h5-trained · mean of steps 1–10

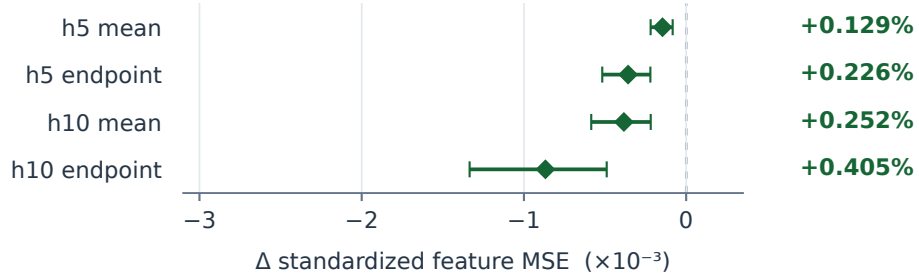
Reduction



b Compare equally h10-trained models

ShiftWM (ours) – Framewise

Reduction



Δ standardized feature MSE ($\times 10^{-3}$)

← Lower error · bars: paired 95% CI

Secondary: ours, matched h5 mean -0.116% (CI includes zero)

Figure 1: Horizon matching on real DROID recordings (original validation only). (a) For each unchanged mode, h10-trained minus original h5-trained standardized feature MSE, averaged over steps 1–10 on identical h10-eligible windows. Training and checkpoint-selection horizons change together. (b) ShiftWM (ours; factorized) minus equally h10-trained Framewise, for all-query mean and endpoint errors at h5 and h10. Dots and bars show signed MSE differences and paired 95% recording-session $\times$ training-seed bootstrap intervals (10,000 draws; seed 5198010). Negative differences favor the first model. Bold green percentages are relative point reductions, not confidence intervals. Both panels use the same absolute-error axis. There are 141 eligible episodes from 59 sessions and 3 model seeds; standard h5 has 1,772 windows, while h10 and matched h5 prefixes have 1,631. Windows are averaged within episodes, then episodes and seeds equally. All 12 runs completed 30 epochs; all selected epoch 1 by the fixed window-weighted validation criterion. The unfavorable secondary matched h5-prefix mean for factorized is -0.116% relative reduction, a small increase in error; new-minus-original MSE difference +0.0001264, paired 95% CI [-0.0002152, +0.0005941]. All 20 registered contrasts are retained in the companion evidence ledger. These are exploratory, multiplicity-unadjusted validation comparisons, not independent test or state-of-the-art claims.